ACM/IDS 104 APPLIED LINEAR ALGEBRA
PROBLEM SET 2: SOLUTIONS

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Problem 1. (10 POINTS) SUBSPACES
Which of the following subsets $W \subset V$ are *subspaces* of the vector spaces V? Explain why.

(a) (1 point) $V = \mathbb{M}_{n\times n}$ and W is a subset of all singular matrices, $W = \{A \,|\, \det A = 0\}$.

(b) (1 point) The *trace* of a square matrix $A \in V = \mathbb{M}_{n\times n}$ is the sum of its diagonal entries:

$$\mathrm{tr}A = \sum_{i=1}^{n} a_{ii}. \tag{1}$$

Let W be a subset of all trace zero matrices, $W = \{A \mid \mathrm{tr}A = 0\}$.

(c) (1 point) Let $V = C([0,1])$ be the vector space of all continuous functions on the interval $[0,1]$, and $W = \{f(x) \in V \mid f(0)f(1) = 1\}$.

(d) (1 points) The vector space V is again $C([0,1])$. The subset $W \subset V$ consists of all functions satisfying the following condition:

$$f\left(\frac{1}{2}\right) = \int_0^1 f(t)dt. \tag{2}$$

(e) (1 point) A *planar vector field* is a function which assigns a vector

$$v(x,y) = \begin{bmatrix} v_1(x,y) \\ v_2(x,y) \end{bmatrix} \tag{3}$$

to each point $(x,y) \in \mathbb{R}^2$. Let V be the vector space of all planar vector fields. A planar vector field is called *incompressible* if it has zero divergence:

$$\nabla \cdot v = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} = 0. \tag{4}$$

Let $W \subset V$ be the subset of all incompressible vector fields.

(f) (2 points) Let V_1 and V_2 be subspaces of a vector space V. Is the union $V_1 \cup V_2$ necessarily a subspace of V?

(g) (3 points) Let V_1 and V_2 be subspaces of a vector space V. Is the intersection $V_1 \cap V_2$ necessarily a subspace of V?

Solution:

(a) Consider the following two matrices:

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \in W \qquad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \in W. \tag{5}$$

Then

$$A + B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \notin W, \tag{6}$$

since $\det(A+B) = 1$. Therefore, W is not a subspace, since it is not closed under addition.

(b) Let's show that for any trace zero matrices A and B and any $\alpha, \beta \in \mathbb{R}$, the linear combination $\alpha A + \beta B$ also has zero trace. This will prove that W is a subspace. Indeed:

$$\mathrm{tr}(\alpha A + \beta B) = \mathrm{tr}(\alpha A) + \mathrm{tr}(\beta B) = \alpha \mathrm{tr}A + \beta \mathrm{tr}B = 0. \tag{7}$$

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(c) Any subspace is a vector space. Any vector space has a zero vector. A zero vector in $V = C([0,1])$ is the zero function $f(x) \equiv 0$. $W \subset V$ is not a subspace since it does not contain $f(x) \equiv 0$.

(d) Let $f, g \in W$ and $\alpha, \beta \in \mathbb{R}$. Let's check that $h = \alpha f + \beta g \in W$:

$$\begin{aligned} h\left(\frac{1}{2}\right) &= (\alpha f + \beta g)\left(\frac{1}{2}\right) = \alpha f\left(\frac{1}{2}\right) + \beta g\left(\frac{1}{2}\right) \\ &= \alpha \int_0^1 f(t)dt + \beta \int_0^1 g(t)dt = \int_0^1 \alpha f(t) + \beta g(t)dt = \int_0^1 h(t)dt. \end{aligned} \tag{8}$$

(e) Let v and w be two incompressible planar vector fields. Let's show that so is $\alpha v + \beta w$:

$$\nabla \cdot (\alpha v + \beta w) = \alpha \nabla v + \beta \nabla w = 0. \tag{9}$$

(f) No, not necessarily. Consider, for instance, the vector space $V = \mathbb{R}^2$ and two subspaces: V_1 is the line $x = 0$ and V_2 is the line $y = 0$. Their union is not a subspace, since it is not closed under addition:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin V_1 \cup V_2. \tag{10}$$

(g) Yes, $V_1 \cap V_2$ is necessarily a subspace of V. This follows from Theorem 1 on page 10 Lecture Notes. Indeed, first, $V_1 \cap V_2$ is nonempty, since the zero vector belongs to both V_1 and V_2. Second, $V_1 \cap V_2$ is closed under vector addition:

$$v, u \in V_1 \cap V_2 \;\Rightarrow\; (v+u) \in V_1, \;\; (v+u) \in V_2 \;\Rightarrow\; (v+u) \in V_1 \cap V_2. \tag{11}$$

Finally, $V_1 \cap V_2$ is closed under scalar multiplication:

$$\alpha \in \mathbb{R}, \; v \in V_1 \cap V_2 \;\Rightarrow\; \alpha v \in V_1, \;\; \alpha v \in V_2 \;\Rightarrow\; \alpha v \in V_1 \cap V_2. \tag{12}$$

Problem 2. (10 POINTS) POLYNOMIALS AND EXPONENTIALS

Let $\mathcal{P}^{(2)}$ be the vector space of quadratic polynomials. Let

$$p_1(x) = x^2 - 3, \quad p_2(x) = 2 - x, \quad p_3(x) = (x+1)^2. \tag{13}$$

(a) (2 points) Determine whether p_1, p_2 and p_3 are linearly independent.

(b) (2 points) Do they span $\mathcal{P}^{(2)}$?

(c) (3 points) Do they form a basis of $\mathcal{P}^{(2)}$? If yes, find the coordinates of $q(x) = 1$ in that basis.

Now consider the following functions:

$$f_1(x) = e^x, \quad f_2(x) = xe^x, \quad f_3(x) = x^2 e^x. \tag{14}$$

(d) (3 points) Determine whether f_1, f_2 and f_3 are linearly independent.

Solution:

(a) Let' consider a linear combination $\alpha_1 p_1(x) + \alpha_2 p_2(x) + \alpha_3 p_3(x)$ and try to find coefficients α_1, α_2, and α_3 (not all zero) such that

$$\alpha_1 p_1(x) + \alpha_2 p_2(x) + \alpha_3 p_3(x) = 0 \quad \text{for all } x. \tag{15}$$

If such α's exist, then the polynomials are linearly dependent, otherwise they are not. After some arithmetics, (15) can be written as follows:

$$(\alpha_1 + \alpha_3)x^2 + (2\alpha_3 - \alpha_2)x + (\alpha_3 + 2\alpha_2 - 3\alpha_1) = 0. \tag{16}$$

The left-hand-side is identically zero if and only if

$$\begin{aligned} &\alpha_1 + \alpha_3 = 0, \\ &2\alpha_3 - \alpha_2 = 0, \\ &\alpha_3 + 2\alpha_2 - 3\alpha_1 = 0. \end{aligned} \tag{17}$$

It is easy to check that this system has only trivial solution $\alpha_1 = \alpha_2 = \alpha_3 = 0$. Therefore, p_1, p_2 and p_3 are linearly independent.

Alternatively, to prove linear independence of p_1, p_2, p_3, we can use the results of Practice Problem 3B. The Wronskian matrix (the matrix of derivatives) is

$$W(x) = \begin{bmatrix} p_1(x) & p_2(x) & p_3(x) \\ p_1'(x) & p_2'(x) & p_3'(x) \\ p_1''(x) & p_2''(x) & p_3''(x) \end{bmatrix} = \begin{bmatrix} x^2-3 & 2-x & (x+1)^2 \\ 2x & -1 & 2(x+1) \\ 2 & 0 & 2 \end{bmatrix} \tag{18}$$

The Wronskian at $x = -1$ is

$$\det W(-1) = \det \begin{bmatrix} -2 & 3 & 0 \\ -2 & -1 & 0 \\ 2 & 0 & 2 \end{bmatrix} = 2 \det \begin{bmatrix} -2 & 3 \\ -2 & -1 \end{bmatrix} = 16 \neq 0. \tag{19}$$

Therefore, the polynomials are linearly independent (for details, see Practice Problem 3B).

(b) Since $\dim \mathcal{P}^{(2)} = 3$ (recall that $1, x, x^2$ form a basis of $\mathcal{P}^{(2)}$) and p_1, p_2, p_3 are linearly independent, they span $\mathcal{P}^{(2)}$ (this follows directly from Theorem 5 on page 15 of Lecture Notes).

(c) Yes, they form a basis of $\mathcal{P}^{(2)}$ because they are linearly independent and span $\mathcal{P}^{(2)}$. To find the coordinates of $q(x) = 1$ in basis p_1, p_2, p_3, we need to solve the following equation for α's:

$$\alpha_1 p_1(x) + \alpha_2 p_2(x) + \alpha_3 p_3(x) = 1 \quad \text{for all } x. \tag{20}$$

This leads to the linear system

$$\begin{aligned} &\alpha_1 + \alpha_3 = 0, \\ &2\alpha_3 - \alpha_2 = 0, \\ &\alpha_3 + 2\alpha_2 - 3\alpha_1 = 1, \end{aligned} \tag{21}$$

Which has the unique solution $\alpha_1 = -1/8$, $\alpha_2 = 1/4$, and $\alpha_3 = 1/8$.

(d) Let's use the results of Practice Problem 3B. The Wronskian matrix of f_1, f_2, and f_3 is

$$W(x) = \begin{bmatrix} f_1(x) & f_2(x) & f_3(x) \\ f_1'(x) & f_2'(x) & f_3'(x) \\ f_1''(x) & f_2''(x) & f_3''(x) \end{bmatrix} = \begin{bmatrix} e^x & xe^x & x^2e^x \\ e^x & e^x + xe^x & 2xe^x + x^2e^x \\ e^x & 2e^x + xe^x & 2e^x + 4xe^x + x^2e^x \end{bmatrix} \tag{22}$$

The Wronskian at $x = 0$ is

$$\det W(0) = \det \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 2 \end{bmatrix} = 2 \neq 0. \tag{23}$$

Therefore, the functions are linearly independent (for details, see Practice Problem 3B).

Problem 3. (10 POINTS) FIBONACCI SEQUENCES

When learning abstract concepts, such as vector spaces, it is handy to have some weird examples at hand that illustrate the concept. These examples will a) develop you abstract thinking and b) motivate you to look for vector spaces in other examples that you will encounter in your study or research[1]. In lectures, we discussed one such example: $(Q, \oplus, *)$. Here is another one.

The Fibonacci sequence is a famous[2] sequence in mathematics and engineering:

$$\mathfrak{f}^* = (1, 1, 2, 3, 5, 8, 13, \ldots). \tag{24}$$

The first two numbers of the sequence are $x_1 = x_2 = 1$ and each subsequent number is the sum of the previous two:

$$x_n = x_{n-1} + x_{n-2}, \quad n = 3, \ldots \tag{25}$$

[1]And as soon as you identify that a set is actually a vector space, you can employ the powerful machinery of linear algebra to attack the underlying problem.

[2]Read `https://en.wikipedia.org/wiki/Fibonacci_number` for information and watch `https://youtu.be/O2wU-HT7FiM` for inspiration.

The sequence $\mathfrak{f}^*$ in (24) is the original Fibonacci sequence. Let us define a generalized Fibanaci sequence $\mathfrak{f}$ as follows:

$$\begin{aligned} &\mathfrak{f} = (x_1, x_2, x_3, \ldots) \\ &x_1 \text{ and } x_2 \text{ are arbitrary real numbers (not necessarily 1)} \\ &x_n = x_{n-1} + x_{n-2}, \quad n = 3, \ldots \end{aligned} \tag{26}$$

That is, in a generalized Fibonacci sequence we can start from any first two numbers. Let $\mathcal{F}$ denote the set of all generalized Fibonacci sequences. Let's define two natural operations on $\mathcal{F}$:

$$\begin{aligned} &\text{Addition: } \mathfrak{f}_1 = (x_1, x_2 \ldots), \;\; \mathfrak{f}_2 = (y_1, y_2, \ldots), \;\; \mathfrak{f}_1 + \mathfrak{f}_2 = (x_1 + y_1, x_2 + y_2 \ldots) \\ &\text{Scalar multiplication: } \mathfrak{f} = (x_1, x_2, \ldots), \;\; \alpha \in \mathbb{R}, \;\; \alpha\mathfrak{f} = (\alpha x_1, \alpha x_2, \ldots). \end{aligned} \tag{27}$$

(a) (4 points) Show that the set of generalized Fibonacci sequences $\mathcal{F}$ is a vector space.
(b) (4 points) Find its dimension and a basis.
(c) (2 points) Find the coordinates of the original sequence $\mathfrak{f}^*$ in that basis.

Solution:

(a) Since $\mathcal{F}$ is a subset of the vector space of infinite sequences of real numbers, all we need to show is that $\mathcal{F}$ is closed under the vector space operations "+" and "·". Let $\mathfrak{f}_1 = (x_1, x_2 \ldots) \in \mathcal{F}$, $\mathfrak{f}_2 = (y_1, y_2 \ldots) \in \mathcal{F}$, $\alpha, \beta \in \mathbb{R}$ and $\mathfrak{f} = \alpha\mathfrak{f}_1 + \beta\mathfrak{f}_2 = (z_1, z_2, \ldots)$. Let's check that $\mathfrak{f} \in \mathcal{F}$, that is $z_n = z_{n-1} + z_{n-2}$:

$$\begin{aligned} z_n &= \alpha x_n + \beta y_n = \alpha(x_{n-1} + x_{n-2}) + \beta(y_{n-1} + y_{n-2}) \\ &= (\alpha x_{n-1} + \beta y_{n-1}) + (\alpha x_{n-2} + \beta y_{n-2}) = z_{n-1} + z_{n-2}. \end{aligned} \tag{28}$$

(b) Any point $\mathfrak{f} \in \mathcal{F}$ is completely determined by its first two components. Moreover, the vector operations that we defined on $\mathcal{F}$ are the same as standard operations on vectors. This suggests that $\mathcal{F}$ is essentially the same[3] as $\mathbb{R}^2$. The following two Fibonacci sequences form a basis of $\mathcal{F}$:

$$\begin{aligned} \mathfrak{f}_1 &= (1, 0, 1, 1, 2, 3, 5, \ldots), \\ \mathfrak{f}_2 &= (0, 1, 1, 2, 3, 5, 8, \ldots). \end{aligned} \tag{29}$$

This makes $\dim \mathcal{F} = 2$.

(c) Since $\mathfrak{f}^* = \mathfrak{f}_1 + \mathfrak{f}_2$, the coordinates of $\mathfrak{f}^*$ are $(1, 1)$.

Problem 4. (10 points) Fundamental Matrix Subspaces

Let's start with a low-dimensional example. Consider the following matrix:

$$A = \begin{bmatrix} 2 & 0 \\ 2 & 2 \\ 20 & 25 \end{bmatrix} \tag{30}$$

(a) (2 points) Find the dimensions of the kernel, cokernel, image, and coimage of A.
(b) (3 points) Find bases (if they exist) for the four fundamental subspaces of A.

Now let's move on to a higher-dimensional problem.

(c) (5 points) Find bases for the kernel, image, cokernel, and coimage of the following matrix:

$$B = \begin{bmatrix} 1 & 2 & \ldots & n \\ n+1 & n+2 & \ldots & 2n \\ \vdots & \vdots & & \vdots \\ n^2 - n + 1 & n^2 - n + 2 & \ldots & n^2 \end{bmatrix} \tag{31}$$

Hint: The proof of the Fundamental Theorem of Linear Algebra (see pages 20–21 of the Lecture Notes) may be helpful for part (c), but it is not necessary.

Solution:

[3] A mathematical way to say it: $\mathcal{F}$ is isomorphic to $\mathbb{R}^2$.

(a) The fundamental theorem of linear algebra tells us that

$$\begin{aligned} \dim\ker A &= n-r \quad \text{and} \quad \dim\operatorname{im}A = r, \\ \dim\operatorname{coker}A &= m-r \quad \text{and} \quad \dim\operatorname{coim}A = r, \end{aligned} \tag{32}$$

where r is the rank of A, m is the number of rows, and n is the number of columns. In our case, $m = 3$, $n = 2$, and $r = 2$, since columns of A are linearly independent. So, the dimensions are

$$\begin{aligned} \dim\ker A &= 0 \quad \text{and} \quad \dim\operatorname{im}A = 2, \\ \dim\operatorname{coker}A &= 1 \quad \text{and} \quad \dim\operatorname{coim}A = 2. \end{aligned} \tag{33}$$

(b) Since $\dim\ker A = 0$, the kernel is trivial (and does not have a basis),

$$\ker A = \{0\}. \tag{34}$$

The image of A is the span of its columns. Since the dimension of image is 2, columns of A form its basis,

$$\operatorname{im}A = \operatorname{span}\left\{ \begin{bmatrix} 2 \\ 2 \\ 20 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 25 \end{bmatrix} \right\} \tag{35}$$

The coimage of A is the span of its rows. Since the dimension of coimage is 2, any two linearly independent rows of A will form a basis, for example

$$\operatorname{coim}A = \operatorname{span}\left\{ \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \end{bmatrix} \right\}. \tag{36}$$

Finally, the cokernel of A is the kernel of A^T, which is

$$A^T = \begin{bmatrix} 2 & 2 & 20 \\ 0 & 2 & 25 \end{bmatrix} \tag{37}$$

Since the dimension of cokernel is 1, it is a span of a nonzero vector $v = (v_1, v_2, v_3)^T$, which is orthogonal to both columns of A or, equivalently, to both rows of A^T:

$$2v_1 + 2v_2 + 20v_3 = 0 \quad \text{and} \quad 2v_2 + 25v_3 = 0. \tag{38}$$

For instance, we can take $v = (5, -25, 2)^T$ and then

$$\operatorname{coker}A = \operatorname{span}\left\{ \begin{bmatrix} 5 \\ -25 \\ 2 \end{bmatrix} \right\}. \tag{39}$$

(c) As we know from Problem 5(a), PS1, the rank of B is two[4]. Therefore, $\dim\operatorname{im}B = \dim\operatorname{coim}B = 2$, and to find bases of $\operatorname{im}B$ and $\operatorname{coim}B$ we need just to find two linearly independent columns and two linearly independent rows of B, respectively. For example, we can take the first two columns and first two rows:

$$\begin{aligned} &\text{A basis of } \operatorname{im}B \text{ is } b_1, b_2, \\ &\text{A basis of } \operatorname{coim}B \text{ is } b^1, b^2. \end{aligned} \tag{40}$$

As discussed in the proof of fundamental theorem of linear algebra, we can construct a basis for the kernel using the row echelon form of B which is (just replace b^2 with $b^2 - (n+1)b^1$)

$$U = \begin{bmatrix} 1 & 2 & 3 & \dots & k & \dots & n \\ 0 & -n & -2n & \dots & -(k-1)n & \dots & -(n-1)n \\ 0 & 0 & 0 & \dots & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \dots & 0 & \dots & 0 \end{bmatrix} \tag{41}$$

[4]Matrix B with $n = 3$ appears also in Practice Problem 7A.

The first two variables, x_1 and x_2, are the basic variables and the remaining $(n-2)$ variables are the free variables. The following collection of $(n-2)$ vectors therefore form a basis of $\ker B$[5]:

$$v_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 2 \\ -3 \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \ldots, v_{n-2} = \begin{bmatrix} n-2 \\ -(n-1) \\ 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix} \tag{42}$$

To find a basis of coker$B$, we can use the row echelon form of B^T, which is

$$W = \begin{bmatrix} 1 & n+1 & 2n+1 & \ldots & (k-1)n+1 & \ldots & (n-1)n+1 \\ 0 & -n & -2n & \ldots & -(k-1)n & \ldots & -(n-1)n \\ 0 & 0 & 0 & \ldots & 0 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \ldots & 0 & \ldots & 0 \end{bmatrix} \tag{43}$$

Again the first two variables are basic and the remaining are free. Setting all free variable to zero except for one, which is 1, leads to the same vectors as in (42). Thus, vectors (42) form a basis of coker$B$.

Problem 5. (10 points) Fundamental Matrix Subspaces: Implementation
Complete Problem 5 in `PS2.mlx`.

Solution: See `PS2solutions.mlx` (uploaded to the Piazza page).

Acknowledgements. A huge thanks to Pavlos Stavrinides for editing my clumsy MATLAB m-files and converting them into neat live scripts!

[5]Compare this basis with the basis of $\ker A$ in Practice Problem 7A (matrix A is matrix B with $n-3$).

ACM/IDS 104 - Problem Set 2 - MATLAB Solutions

Before writing your MATLAB code, it is always good practice to get rid of any leftover variables and figures from previous scripts.

```
clc; clear; close all;
```

Problem 5 (10 points) Fundamental Matrix Subspaces

Your task for this problem is to write a function that takes a matrix `A` as its argument, and outputs four matrices: `K, I, cK` and `cI` where:

- Columns of `K` form a basis of the kernel of `A`. If $\ker A = \{0\}$, then `K` must be a zero vector of the appropriate dimension.
- Columns of `I` form a basis of the image of `A`. If $\mathrm{im} A = \{0\}$, then `I` must be a zero vector of the appropriate dimension.
- Columns of `cK` form a basis of the cokernel of `A`. If $\mathrm{coker} A = \{0\}$, then `cK` must be a zero vector of the appropriate dimension.
- Columns of `cI` form a basis of the coimage of `A`. If $\mathrm{coim} A = \{0\}$, then `cI` must be a zero vector of the appropriate dimension.

Move to the bottom of this livescript to write the function.

Now, let us test our function:

```
A = magic(6); % feel free to define A as you like
[K, I, cK, cI] = subspacer(A); % this is how you call a MATLAB function
disp(K);
```

```
   -0.4714
   -0.4714
    0.2357
    0.4714
    0.4714
   -0.2357
```

```
disp(I);
```

```
   -0.4082    0.5574    0.0456   -0.4182    0.3092
   -0.4082   -0.2312    0.6301   -0.2571   -0.5627
   -0.4082    0.4362    0.2696    0.5391    0.1725
   -0.4082   -0.3954   -0.2422   -0.4590    0.3971
   -0.4082    0.1496   -0.6849    0.0969   -0.5766
   -0.4082   -0.5166   -0.0182    0.4983    0.2604
```

```
disp(cK);
```

```
    0.5000
   -0.0000
   -0.5000
   -0.5000
   -0.0000
    0.5000
```

```
disp(cI);
```

```
   -0.4082    0.6234   -0.3116    0.2495   -0.2511
   -0.4082   -0.6282    0.3425    0.1753   -0.2617
   -0.4082   -0.4014   -0.7732   -0.0621    0.1225
   -0.4082    0.1498    0.2262   -0.4510   -0.5780
   -0.4082    0.1163    0.2996    0.6340    0.3255
   -0.4082    0.1401    0.2166   -0.5457    0.6430
```

START HERE by writing the function:

```
function [K, I, cK, cI] = subspacer(A)
%{
This is the MATLAB function syntax.
-> [K, I, cK, cI] are the outputs of the function.
-> "subspacer" is the name of the function. (you can change that if
                            you wish but make sure you change
                            every function call as well!)
-> A is the argument of the function.
%}
[m, n] = size(A);
r = rank (A);
%{
We start by finding out the dimensions and rank of A.
Let us consider the matrix K. There exist 2 cases:
1) The kernel is trivial i.e. kerA = {0}
2) The kernel is not trivial -> Hint: use null()
Complete the following if/else statement.
%}
if r == n % this condition is done for you
    K = zeros(n, 1);
else
    K = null(A);
end
%{
Now, let us consider the matrix cK.
As above, there exist 2 cases. Remember, you can use ' to
transpose a matrix.
Complete the following if/else statement.
%}
if r == m % fill in the condition
    cK = zeros(m, 1);
else
    cK = null(A');
end
%{
For the image I and coimage cI, there exists only 1 condition
we must test, and that is if rankA = 0. With this in mind,
complete the following if/else statement.
-> Hint: orth() is useful here.
%}
```

```
if r == 0
    I = zeros(m, 1);
    cI = zeros(n ,1);
else
    I = orth(A);
    cI = orth(A');
end
end
```